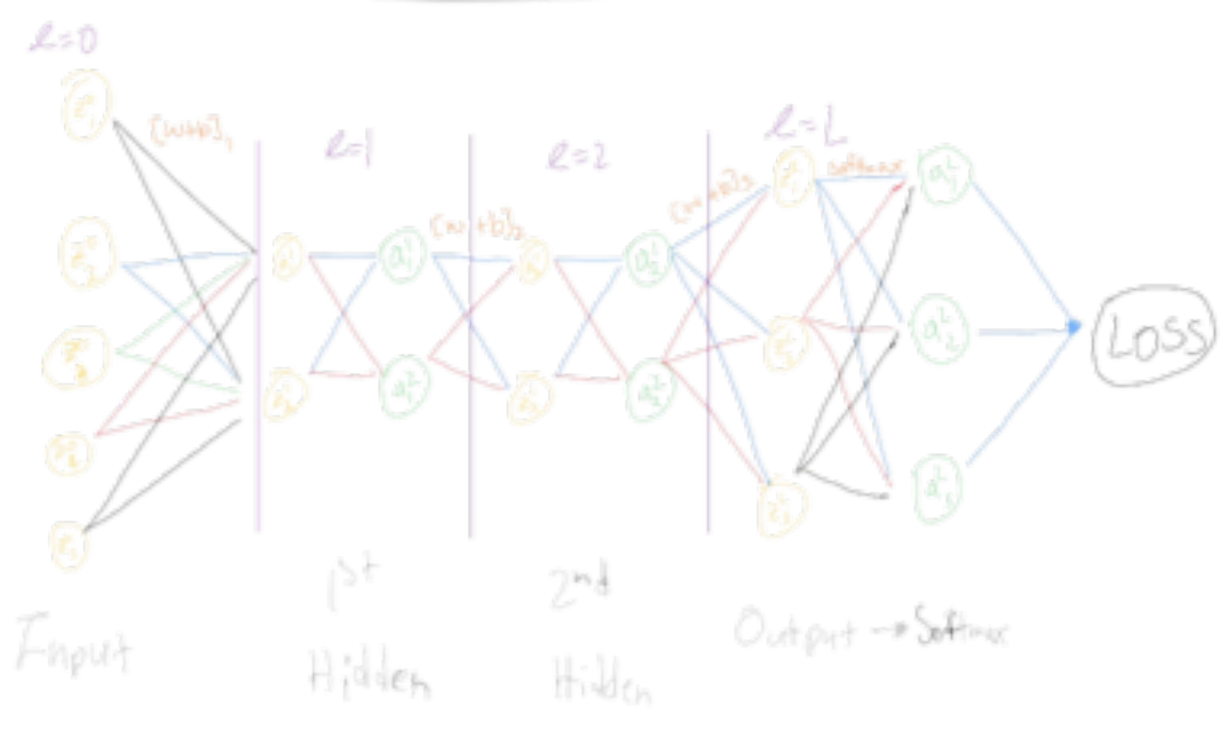
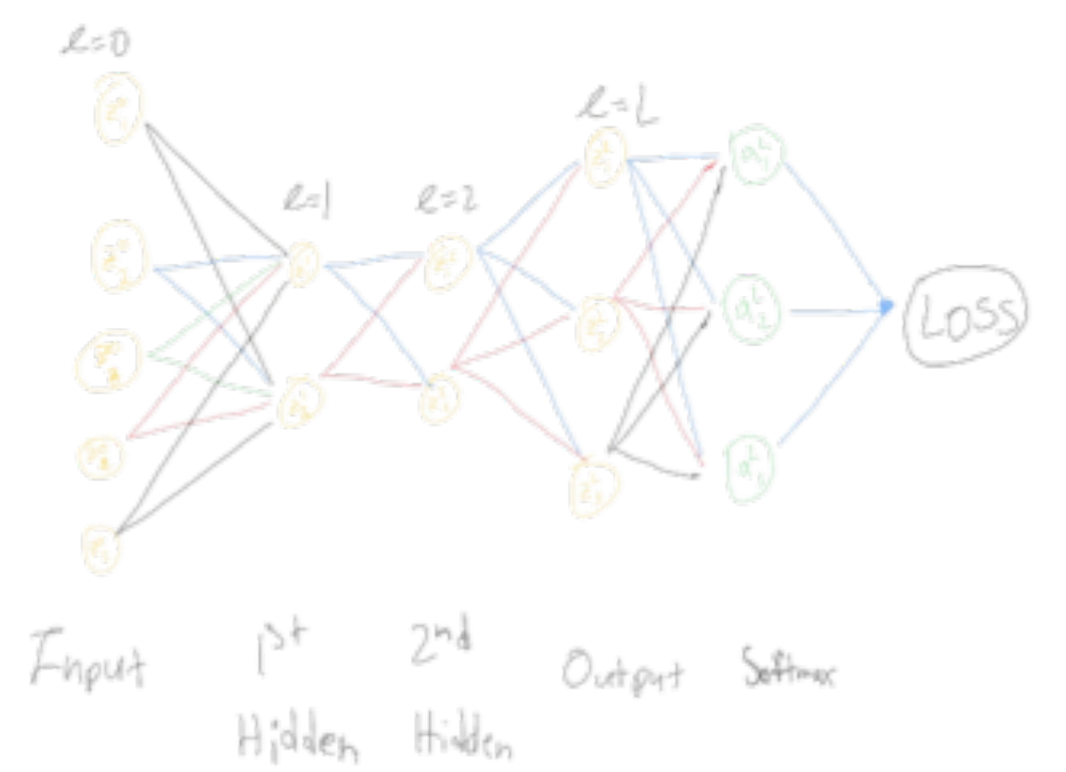
Backpropagation

Example network



VI:



Notation & Set-Up

- ℓ: denotes specific layer #, with $\ell=1$ for 1st hidden layer & L = output layer
- W_{jk}^{ℓ}: Weight connecting neuron k in layer $\ell-1$ to neuron j in layer ℓ
- b_j^{ℓ}: Bias for neuron j in layer ℓ
- z_j^{ℓ}: Net weighted input for neuron j in layer ℓ
- a_j^{ℓ}: Activation for neuron j in layer ℓ (where $a_j^{\ell} = \sigma(z_j^{\ell})$). For input layer, $a^0 = x$
- δ_j^{ℓ}: Error gradient for neuron j in layer ℓ, defined as $\frac{\partial L}{\partial z_j^{\ell}}$
- N^{ℓ}: # of neurons in layer ℓ

Error Gradient for Output Layer

Multiclass - Classification Loss Function:

1.)

a.) $\text{Loss } (L) = -\sum_{c=1}^{C} y_c \cdot \ln(a_c^L)$

where y_c = one-hot encoded ground truth at class c

and a_c^L = Softmax activation of neuron/class c in layer L

b.) $a_i^L = \frac{e^{(z_i^L)}}{\sum_{m=1}^{C} e^{(z_m^L)}} \Rightarrow$ (Softmax)

2.) $\delta_j^L = \frac{\partial L}{\partial z_j^L} = \_\_\_\_$ (completely foiled out)

But, more succintly:

$$\delta_j^L = \sum_{c=1}^{C} \frac{\partial L}{\partial a_c^L} \cdot \frac{\partial a_c^L}{\partial z_j^L},$$

where there are 2 cases for when $j=c$ & $j \neq c$. This makes sense because the softmax activation a_c^L is influenced by its respective weighted neuron z_c^L more than a different neuron $z_{j\neq c}^L$ because the numerator of a_c^L contains a function of z_c^L, but not of $z_{j\neq c}^L$.

$$\delta_j^L = \left[\frac{\partial L}{\partial a_j^L} \cdot \frac{\partial a_j^L}{\partial z_j^L}\right] + \left[\sum_{c\neq j}^{C} \frac{\partial L}{\partial a_c^L} \cdot \frac{\partial a_c^L}{\partial z_j^L}\right]$$

$$= \left[\frac{-y_j}{a_j^L} \cdot \left(a_j^L \cdot (1-a_j^L)\right)\right] + \left[\sum_{c\neq j}^{C} \frac{-y_c}{a_c^L} \cdot (-a_c^L \cdot a_j^L)\right]$$

$$= \left[-y_j \cdot (1-a_j^L)\right] + \left[\sum_{c\neq j}^{C} y_c \cdot a_j^L\right]$$

$$= -y_j + \underbrace{y_j a_j^L + \sum_{c\neq j}^{C} y_c \cdot a_j^L}_{=\text{ Sum of all } c \text{ from } 1 \text{ to } C} = -y_j + \sum_{c=1}^{C} y_c a_j^L$$

$$= -y_j + a_j^L \cdot \underbrace{\left(\sum_{c=1}^{C} y_c\right)}_{=1 \text{ (Sum of one-hot encoded vector } =1)} = \boxed{a_j - y_j}$$

For every neuron j in the output layer, $\delta_j^L = a_j - y_j$. Thus, the error gradient vector:

$$\delta^L = \hat{a}_j - \hat{y}_j$$

Foiled Out

$$\text{loss} = -\sum_{i=1}^{C} y_i \cdot \ln(a_i^L)$$

$$= -\left[(y_1 \ln(a_1^L)) + (y_2 \ln(a_2^L)) + (y_3 \ln(a_3^L))\right]$$

$$\frac{\partial L}{\partial z_1^L} = \left(\frac{\partial L}{\partial a_1^L} \cdot \frac{\partial a_1^L}{\partial z_1^L}\right) + \left(\frac{\partial L}{\partial a_2^L} \cdot \frac{\partial a_2^L}{\partial z_1^L}\right) + \left(\frac{\partial L}{\partial a_3^L} \cdot \frac{\partial a_3^L}{\partial z_1^L}\right)$$

$$= \left[-\frac{y_1}{a_1^L} \cdot \overbrace{\left(\frac{\partial}{\partial z_1^L} \sigma(a_1^L)\right)}^{a}\right] + \left[-\frac{y_2}{a_2^L} \cdot \overbrace{\left(\frac{\partial}{\partial z_1^L} \sigma(a_2^L)\right)}^{b}\right] + \left[-\frac{y_3}{a_3^L} \cdot \overbrace{\left(\frac{\partial}{\partial z_1^L} \sigma(a_3^L)\right)}^{c}\right]$$

a.) $\frac{\partial}{\partial z_1^L} \sigma(a_1^L) = \frac{\left(e^{(z_1^L)} \cdot \sum_{m=1}^{C} e^{(z_m^L)}\right) - \left(e^{(z_1^L)} \cdot e^{(z_1^L)}\right)}{\left(\sum_{m=1}^{C} e^{(z_m^L)}\right)^2} = \left(\frac{e^{(z_1^L)}}{\sum_{m=1}^{C} e^{(z_m^L)}}\right) - \left(\frac{e^{(z_1^L)}}{\sum_{m=1}^{C} e^{(z_m^L)}}\right)^2$

$f = e^{(z_1^L)}$ $\quad = \frac{u'v - uv'}{v^2}$

$g = \sum_{m=1}^{C} e^{(z_m^L)}$

$\frac{\partial}{\partial z_1^L} f = e^{(z_1^L)}$

$\frac{\partial}{\partial z_1^L} g = e^{(z_1^L)}$

$$= a_1^L - (a_1^L)^2$$

$$= \boxed{a_1^L(1-a_1^L)}$$

Identical for arbitrary $z_{i\neq c}$

b.) $\frac{\partial}{\partial z_1^L} \sigma(a_2^L) = \frac{\cancel{0} - \left(e^{(z_2^L)} \cdot e^{(z_1^L)}\right)}{\left(\sum_{m=1}^{C} e^{(z_m^L)}\right)^2} = -\left[\frac{e^{(z_2^L)}}{\sum_{m=1}^{C} e^{(z_m^L)}} \cdot \frac{e^{(z_1^L)}}{\sum_{m=1}^{C} e^{(z_m^L)}}\right]$

$f = e^{\overbrace{(z_2^L)}^{a_2}}$

$g = \sum_{m=1}^{C} e^{(z_m^L)}$

$\frac{\partial}{\partial z_1^L} f = 0$

$\frac{\partial}{\partial z_1^L} g = e^{(z_1^L)}$

$$= \boxed{-(a_2^L \cdot a_1^L)}$$

c.) $\frac{\partial}{\partial z_1^L} \sigma(a_3^L) = \frac{\cancel{0} - \left(e^{(z_3^L)} \cdot e^{(z_1^L)}\right)}{\left(\sum_{m=1}^{C} e^{(z_m^L)}\right)^2} = -\left[\frac{e^{(z_3^L)}}{\sum_{m=1}^{C} e^{(z_m^L)}} \cdot \frac{e^{(z_1^L)}}{\sum_{m=1}^{C} e^{(z_m^L)}}\right]$

$f = e^{\overbrace{(z_3^L)}^{a_2}}$

$g = \sum_{m=1}^{C} e^{(z_m^L)}$

$\frac{\partial}{\partial z_1^L} f = 0$

$\frac{\partial}{\partial z_1^L} g = e^{(z_1^L)}$

$$= \boxed{-(a_3^L \cdot a_1^L)}$$

$$\frac{\partial L}{\partial z_1^2} = \left(\frac{\partial L}{\partial a_1^L} \cdot \left(-a_1^L(1-a_1^L)\right)\right) + \left(\frac{\partial L}{\partial a_2^L} \cdot \left(-(a_2^L \cdot a_1^L)\right)\right) + \left(\frac{\partial L}{\partial a_3^L} \cdot \left(-(a_3^L \cdot a_1^L)\right)\right)$$

$$= \left(\frac{-y_1}{\cancel{a_1^L}} \cdot \left(\cancel{a_1^L}(1-a_1^L)\right)\right) + \left(\frac{-y_2}{\cancel{a_2^L}} \cdot \left(-(\cancel{a_2^L} \cdot a_1^L)\right)\right) + \left(\frac{-y_3}{\cancel{a_3^L}} \cdot \left(-(\cancel{a_3^L} \cdot a_1^L)\right)\right)$$

$$= -y_1(1-a_1^L) + y_2 a_1^L + y_3 a_1^L = -y_1 + y_1 a_1^L + y_2 a_1^L + y_3 a_1^L$$

$$= -y_1 + a_1^L \cdot \underbrace{(y_1 + y_2 + y_3)}_{=1} = -y_1 + a_1^L = \boxed{a_1^L - y_1}$$

Partial Derivative of Loss w.r.t. Every Neuron

Formula: $z_j^{\ell} = \left[\sum_{k=1}^{N^{\ell-1}} (w_{jk}^{\ell} \cdot a_k^{\ell-1})\right] + b_j^{\ell}$, Find: $\frac{\partial L}{\partial z_j^{\ell}}$

Start w/ case $\ell = L-1$:

Find $\frac{\partial L}{\partial z_j^{L-1}} = \delta_j^{L-1}$

$$\delta_j^{L-1} = \frac{\partial L}{\partial z_j^{L-1}} = \sum_{k=1}^{N^L} \overbrace{\frac{\partial L}{\partial z_k^L}}^{=\delta_k^L} \cdot \frac{\partial z_k^L}{\partial z_j^{L-1}}$$

$$z_k^L = \left[\sum_{j=1}^{N^{L-1}} (w_{kj}^{L+1} \cdot a_j^{L-1})\right] + b_k^L$$

$$a_j^{L-1} = \sigma(z_j^{L-1}) \rightarrow z_k^L = \left[\sum_{j=1}^{N^{L-1}} (w_{kj}^{L+1} \cdot \sigma(z_j^{L-1}))\right] + b_k^L$$

$$\frac{\partial z_k^L}{\partial z_j^{L-1}} = \left(w_{kj}^{L-1} \cdot \sigma'(z_j^{L-1})\right)$$

$$\delta_j^{L-1} = \sum_{k=1}^{N^L} \delta_k^L \cdot \left(w_{kj}^{L-1} \cdot \sigma'(z_j^{L-1})\right)$$

$$= \sum_{k=1}^{N^L} \delta_k^L \cdot w_{kj}^{L-1} \cdot \sigma'(z_j^{L-1}) = \left(\sigma'(z_j^{L-1})\right) \cdot \left(\sum_{k=1}^{N^L} \delta_k^L \cdot w_{kj}^{L-1}\right)$$

$$\delta^{L-1} = [\delta_{j=1}^{L-1}, \delta_{j=2}^{L-1}, \delta_{j=3}^{L-1}, \ldots \delta_{j=N^{L-1}}^{L-1}] \Rightarrow \text{vector of each } \delta_j^{L-1}$$

$\left(\sum_{k=1}^{N^L} \delta_k^L \cdot w_{kj}^{L-1}\right)$:

1.) Shape (Post-Sum) = $[1 \times 1]$ (because it's a Σ, & also it = $\sigma'(z_j^{L-1})$ shape)

2.) ☆ Assume W is of shape [output x input] = $[k \times j]$

$\Rightarrow [1 \times k] \cdot [k \times 1] = [1 \times 1]$

$\Rightarrow$ Either: $(\delta^L)(W_j^L)$ OR $(W_j^L)^T(\delta^L)$ ✓

$\Rightarrow$ When iterating over all values for j, we can simplify to:

$$\delta^{L-1} = \left((W^L)^T \delta^L\right) \odot \sigma'(\vec{z}^{L-1})$$

, where $\odot$ is the element-wise (Hadamard) product (builds the vector δ^{L-1})

Generalize for all layers:

$$\delta^{\ell-1} = \left((W^{\ell})^T \delta^{\ell}\right) \odot \sigma'(\vec{z}^{\ell-1})$$

Derivative of Loss w.r.t. weights & biases

$$z_j^{\ell} = \left[\sum_{k=1}^{N^{\ell}} (w_{jk}^{\ell} \cdot a_k^{\ell-1})\right] + b_j^{\ell}$$

$$\frac{\partial L}{\partial w_{jk}^{\ell}} = \frac{\partial L}{\partial z_j^{\ell}} \cdot \frac{\partial z_j^{\ell}}{\partial w_{jk}^{\ell}} = \delta_j^{\ell} \cdot a_k^{\ell-1}$$

$$\frac{\partial L}{\partial b_j^{\ell}} = \frac{\partial L}{\partial z_j^{\ell}} \cdot \frac{\partial z_j^{\ell}}{\partial b_j^{\ell}} = \delta_j$$

Updating weights & biases

α = Learning Rate

$$w_{jk}^{\ell} = w_{jk}^{\ell} - \left[\alpha \cdot \left(\delta_j^{\ell} \cdot a_k^{\ell-1}\right)\right]$$

$$b_j^{\ell} = b_j^{\ell} - \left[\alpha \cdot \left(\delta_j^{\ell}\right)\right]$$

↓ Matrix Multiplication

$$W^{\ell} = W^{\ell} - \left[\alpha \cdot \left(\hat{\delta}^{\ell} \cdot \hat{a}^{\ell-1}\right)\right]$$

$$b^{\ell} = b^{\ell} - \left[\alpha \cdot \delta^{\ell}\right]$$